Homework 3

Hardware in the Loop

How to submit

Write one PDF. It describes the connector, the MCP server, which agent holds the hardware tools, and the hardware-in-the-loop turn. It includes an architecture diagram of the system so far and sequence diagrams of the runs. It also records the use cases: the request, the tool calls, and what the instrument did. The PDF gives the GitHub link to the code.

Push the code to a GitHub repository. The repository may be the one you used for Homework 1 and Homework 2, with this work added. Put the demo video in that repository as well, or attach it to the email together with the PDF.

Send one email to aramsargsyan@ysu.am with that PDF attached.

Task 1. Hardware in the loop

Homework 1, Task 6, is the LangGraph system of four roles. Homework 2 adds a human stop to that system, and a simulated bench. This homework puts a real instrument on the same system. The instrument is reached through its own software, the controls are exposed as tools, and one agent is the role that calls them.

Problem.

(a) The connector. 4 points

Establish the connection to the real hardware through the software that instrument provides: its SDK, its API, or the other interface it documents. This connector is the code that sends a command and reads a measurement back. Name the instrument and the interface you used.

(b) The MCP server. 4 points

Build an MCP server that exposes the controls of that hardware as tools. A tool is one action the instrument can perform, or one reading it can return. The server carries out a tool call by using the connector from (a).

(c) One agent holds the tools. 4 points

Give the MCP server to one agent in the system from Homework 1 and Homework 2. That agent is the one that calls the hardware. The decision to call a tool is a model call, made from the dialogue.

(d) Hardware in the loop. 4 points

Add a hardware turn in the same spirit as the human stop from Homework 2. When the next step needs the instrument, the graph gives the turn to the agent that holds the MCP tools. That agent calls a tool, the instrument acts, and the result is written into the dialogue. The following turn reads that result before it chooses again. The loop is: decide, command the hardware, receive the measurement, decide again.

(e) Diagrams of the system so far. 4 points

In the PDF, give an architecture diagram of the whole system as it stands after this homework: the roles from Homework 1, the human stop and the simulated bench from Homework 2, and the connector, the MCP server, and the hardware turn from this homework. Give sequence diagrams of the runs, showing the order of messages among the agents, the human, the MCP tools, and the instrument.

(f) Use cases and the demo. 6 points

Test the system on several use cases. For each one, the PDF states the request you gave, the tool calls the agent made, and what the hardware did in response. **Screen record a video** that shows the agent making a decision, calling the hardware through MCP, and the instrument responding. Add the video to the GitHub repository, or send it by email with the PDF.

Push the code for this task to a GitHub repository and put the link in the PDF.